

A four-letter infinite word avoiding weak abelian cubes

Abstract

We give an explicit infinite word over $\{0, 1, 2, 3\}$ with no factor xyz in which the three nonempty blocks have the same normalized Parikh vector. The word is obtained by coding a fixed point of a substitution on five auxiliary letters. The proof uses a ternary recursion and the quadratic form $X^2 - 3Y^2$: at equal endpoint states, the 3-adic valuation of the form evaluated on a displacement equals the valuation of the corresponding difference of indices. This excludes four collinear return vertices and hence excludes every weak abelian cube.

All indices start at 0, and $\mathbb{N} = \{0, 1, 2, \dots\}$. For a finite word v over $\Sigma = \{0, 1, 2, 3\}$, let $|v|$ denote its length, let $|v|_j$ denote the number of occurrences of the letter j , and define its Parikh vector by

$$\psi(v) = (|v|_0, |v|_1, |v|_2, |v|_3)^\top \in \mathbb{Z}_{\geq 0}^4.$$

A *weak abelian cube* is a concatenation xyz of three nonempty finite words such that

$$\frac{\psi(x)}{|x|} = \frac{\psi(y)}{|y|} = \frac{\psi(z)}{|z|}.$$

The three block lengths need not be equal. A factor of an infinite word means a finite contiguous block.

The construction below is a ternary adaptation of the valuation method used by Cambie and Kalviainen [1] for a finite-step lattice walk with no three collinear vertices. The argument here is self-contained; in particular, no finite search is used as a premise.

Theorem 1. *Let $\mathcal{A} = \{A, B, C, D, E\}$. Define a substitution $\tau : \mathcal{A}^* \rightarrow \mathcal{A}^*$ and a letter-to-letter coding $h : \mathcal{A}^* \rightarrow \Sigma^*$ by*

a	A	B	C	D	E
$\tau(a)$	AB	$AACA$	ADE	$AACCE$	$ADCCA$
$h(a)$	0	1	2	3	3

and extend both maps to words by concatenation. Then

$$\boxed{\mathbf{w} = h(\tau^\infty(A))}$$

is an infinite word over Σ containing no weak abelian cube.

The limit $\tau^\infty(A)$ is well defined: $\tau(A)$ starts with A , so $\tau^k(A)$ is a prefix of $\tau^{k+1}(A)$, and all images have length at least 2, so these prefix lengths tend to infinity. Since h maps every auxiliary letter to one letter, its image is also infinite. For reference, the word begins

$$\mathbf{w} = 0100200101033010100200100200100223 \cdots .$$

All four output letters occur: A, B, C occur in $\tau^2(A) = ABAACA$, and D, E occur in $\tau^3(A)$.

We first identify the blocks encoded by the auxiliary letters. Define another substitution, this time on Σ , by

$$\sigma(0) = 010, \quad \sigma(1) = 232, \quad \sigma(2) = 101, \quad \sigma(3) = 323, \quad (1)$$

and put $\mathbf{t} = \sigma^\infty(0)$. This fixed point exists because $\sigma(0)$ starts with 0 and every image has length 3. Define $R : \mathcal{A}^* \rightarrow \Sigma^*$ by the following nonempty images:

$$\begin{array}{c|ccccc} a & A & B & C & D & E \\ \hline R(a) & 01 & 0232 & 013231 & 02323231 & 01323232. \end{array} \quad (2)$$

Lemma 2. *Writing $\mathbf{s} = s_0 s_1 s_2 \cdots = \tau^\infty(A)$, we have*

$$R(\mathbf{s}) = R(s_0)R(s_1)R(s_2) \cdots = \mathbf{t}.$$

Moreover, the starts of these blocks are exactly the positions at which $\mathbf{t}$ has the letter 0.

Proof. The five required substitution identities are as follows. Spaces on the right separate the successive R -images.

$$\begin{aligned} \sigma(R(A)) &= 01 \ 0232 = R(A)R(B), \\ \sigma(R(B)) &= 01 \ 01 \ 013231 \ 01 = R(A)R(A)R(C)R(A), \\ \sigma(R(C)) &= 01 \ 02323231 \ 01323232 = R(A)R(D)R(E), \\ \sigma(R(D)) &= 01 \ 01 \ 013231 \ 013231 \ 01323232 = R(A)R(A)R(C)R(C)R(E), \\ \sigma(R(E)) &= 01 \ 02323231 \ 013231 \ 013231 \ 01 = R(A)R(D)R(C)R(C)R(A). \end{aligned}$$

Thus $\sigma \circ R = R \circ \tau$ on every letter, and hence on every finite word. Induction on k gives

$$R(\tau^k(A)) = \sigma^k(R(A)) = \sigma^k(01).$$

The word on the right starts with $\sigma^k(0)$, whose length is 3^k . Consequently, for every fixed prefix length, these prefixes eventually agree with $\mathbf{t}$. On the left, the nested words converge to $R(\mathbf{s})$. This proves $R(\mathbf{s}) = \mathbf{t}$.

Each of the five words in (2) begins with 0 and has no other occurrence of 0. The concatenation is infinite and each block is nonempty. Its block starts are therefore precisely all the zero positions of $\mathbf{t}$. $\square$

We next associate a lattice walk to $\mathbf{t} = t_0 t_1 t_2 \cdots$. For each $n \in \mathbb{N}$, define (a_n, b_n) from t_n by

$$\begin{array}{c|cccc} t_n & 0 & 1 & 2 & 3 \\ \hline a_n & 1 & 1 & -1 & -1 \\ b_n & 1 & -1 & 1 & -1 \end{array} \quad (3)$$

and define integer prefix sums and vertices by

$$X_n = \sum_{j=0}^{n-1} a_j, \quad Y_n = \sum_{j=0}^{n-1} b_j, \quad P_n = (X_n, Y_n, n) \in \mathbb{Z}^3. \quad (4)$$

An empty sum is zero, so $P_0 = (0, 0, 0)$. Finally, put

$$F(U, V) = U^2 - 3V^2.$$

For a nonzero integer q , $\nu_3(q)$ denotes the largest nonnegative integer e such that 3^e divides q ; the sign of q is irrelevant. Extend this to nonzero rational numbers by $\nu_3(p/q) = \nu_3(p) - \nu_3(q)$. In particular, $\nu_3(uv) = \nu_3(u) + \nu_3(v)$ for nonzero rational u, v .

Lemma 3. *For every $n \in \mathbb{N}$ and $r \in \{0, 1, 2\}$,*

$$a_{3n+r} = b_n, \quad b_{3n+r} = (-1)^r a_n, \quad (5)$$

$$X_{3n+r} = 3Y_n + r b_n, \quad Y_{3n+r} = X_n + \varepsilon_r a_n, \quad (6)$$

where $\varepsilon_0 = 0$, $\varepsilon_1 = 1$, and $\varepsilon_2 = 0$.

Proof. For an input state (a_n, b_n) , the three output states in its σ -image, as read from (1) and (3), are

$$(b_n, a_n), \quad (b_n, -a_n), \quad (b_n, a_n).$$

This proves (5). The sum of their first coordinates is $3b_n$, and the sum of their second coordinates is a_n . Thus $X_{3n} = 3Y_n$ and $Y_{3n} = X_n$. Within the next block, the first r first coordinates sum to $r b_n$, while the first r second coordinates sum to $0, a_n, 0$ for $r = 0, 1, 2$, respectively. Adding these partial sums proves (6). $\square$

Lemma 4 (Equal-state valuation identity). *If $0 \leq m < n$ and $t_m = t_n$, then*

$$F(X_n - X_m, Y_n - Y_m) \neq 0$$

and

$$\nu_3(F(X_n - X_m, Y_n - Y_m)) = \nu_3(n - m). \quad (7)$$

Proof. Write $m = 3u + r$ and $n = 3v + s$, with $r, s \in \{0, 1, 2\}$. We separate the cases $r \neq s$ and $r = s$.

Suppose first that $r \neq s$. The endpoint states agree, so $a_m = a_n$. Equation (5) implies

$$b_u = a_m = a_n = b_v.$$

Consequently, by (6),

$$X_n - X_m = 3(Y_v - Y_u) + sb_v - rb_u \equiv (s - r)a_m \pmod{3}.$$

Here $a_m \in \{-1, 1\}$, and $s - r$ is not divisible by 3. Therefore $X_n - X_m$ is not divisible by 3, and

$$F(X_n - X_m, Y_n - Y_m) \equiv (X_n - X_m)^2 \not\equiv 0 \pmod{3}.$$

The quadratic expression is nonzero and has valuation zero. Also $3 \nmid n - m$, so (7) holds in this case.

Now suppose $r = s$. Equality of the first coordinates of the endpoint states gives $b_u = b_v$, and equality of the second coordinates gives $(-1)^r a_u = (-1)^r a_v$, hence $a_u = a_v$. The correspondence in (3) is one-to-one, so $t_u = t_v$. In particular, the correction terms in (6) cancel, giving

$$X_n - X_m = 3(Y_v - Y_u), \quad Y_n - Y_m = X_v - X_u.$$

It follows that

$$\begin{aligned} F(X_n - X_m, Y_n - Y_m) &= 9(Y_v - Y_u)^2 - 3(X_v - X_u)^2 \\ &= -3F(X_v - X_u, Y_v - Y_u). \end{aligned} \tag{8}$$

At the same time, $n - m = 3(v - u)$, and $u < v$.

Let $e = \nu_3(n - m)$. Apply this second case exactly e times. At each step the endpoint states still agree, the indices remain distinct and ordered, and their difference is divided by 3. We arrive at indices $m' < n'$ such that $t_{m'} = t_{n'}$ and $3 \nmid n' - m'$, with

$$F(X_n - X_m, Y_n - Y_m) = (-3)^e F(X_{n'} - X_{m'}, Y_{n'} - Y_{m'}).$$

The first case shows that the final factor is nonzero and has valuation zero. The original expression is therefore nonzero and has valuation e , as required. $\square$

Proposition 5. *No four distinct vertices in the set*

$$\{P_n : n \in \mathbb{N}, t_n = 0\}$$

are collinear.

Proof. Suppose that $P_{n_0}, P_{n_1}, P_{n_2}, P_{n_3}$ are collinear, where $n_0 < n_1 < n_2 < n_3$ and $t_{n_i} = 0$ for each i . Their third coordinates are distinct, so the line has well-defined rational slopes

$$\alpha = \frac{X_{n_1} - X_{n_0}}{n_1 - n_0}, \quad \beta = \frac{Y_{n_1} - Y_{n_0}}{n_1 - n_0}.$$

For every $0 \leq i < j \leq 3$, collinearity gives

$$X_{n_j} - X_{n_i} = \alpha(n_j - n_i), \quad Y_{n_j} - Y_{n_i} = \beta(n_j - n_i).$$

Put $c = \alpha^2 - 3\beta^2$. Applying Lemma 4 to the pair (n_0, n_1) shows that $c \neq 0$. For every pair (i, j) , the same lemma now gives

$$\begin{aligned}\nu_3(n_j - n_i) &= \nu_3(c(n_j - n_i)^2) \\ &= \nu_3(c) + 2\nu_3(n_j - n_i).\end{aligned}$$

Hence all six positive differences $n_j - n_i$ have the same valuation

$$e = -\nu_3(c) \in \mathbb{Z}_{\geq 0}.$$

The four numbers

$$q_i = \frac{n_i - n_0}{3^e} \quad (0 \leq i \leq 3)$$

are integers. For $i < j$, the difference $q_j - q_i = (n_j - n_i)/3^e$ is not divisible by 3. Thus the four q_i would be pairwise distinct modulo 3. There are only three residue classes modulo 3, a contradiction. $\square$

We can now relate the geometry to the normalized Parikh vectors of $\mathbf{w}$. For a finite word v over Σ , define its displacement by

$$\Delta(v) = (|v|_0 + |v|_1 - |v|_2 - |v|_3, |v|_0 - |v|_1 + |v|_2 - |v|_3, |v|)^T.$$

By (3) and (4), if v is the factor $t_m t_{m+1} \cdots t_{n-1}$, then $\Delta(v) = P_n - P_m$. The return blocks have the following Parikh vectors and displacements:

a	$h(a)$	$\psi(R(a))^T$	$\Delta(R(a))^T$
A	0	(1, 1, 0, 0)	(2, 0, 2)
B	1	(1, 0, 2, 1)	(-2, 2, 4)
C	2	(1, 2, 1, 2)	(0, -2, 6)
D	3	(1, 1, 3, 3)	(-4, 0, 8)
E	3	(1, 1, 3, 3)	(-4, 0, 8)

In particular, the displacement depends only on $h(a)$, including for the two auxiliary letters D, E that share the output letter 3. Define the linear map $L : \mathbb{Q}^4 \rightarrow \mathbb{Q}^3$ by the matrix

$$L = \begin{pmatrix} 2 & -2 & 0 & -4 \\ 0 & 2 & -2 & 0 \\ 2 & 4 & 6 & 8 \end{pmatrix}. \quad (9)$$

Its columns, in output-letter order 0, 1, 2, 3, are the four return displacements above.

Proof of Theorem 1. For $\mathbf{s} = s_0 s_1 s_2 \cdots$, let

$$N_0 = 0, \quad N_k = \sum_{j=0}^{k-1} |R(s_j)| \quad (k \geq 1).$$

By Lemma 2, these are precisely the zero positions of $\mathbf{t}$, in increasing order. The strict increase follows also from the fact that every return block has positive length.

Write $\mathbf{w} = w_0 w_1 w_2 \cdots$, where $w_j = h(s_j)$. Additivity of displacements and the return-block table show that, for all $0 \leq p < q$,

$$P_{N_q} - P_{N_p} = \sum_{j=p}^{q-1} \Delta(R(s_j)) = L\psi(w_p w_{p+1} \cdots w_{q-1}). \quad (10)$$

This holds even though h is not one-to-one, because $R(D)$ and $R(E)$ have equal displacements.

Suppose for a contradiction that a factor of $\mathbf{w}$ is a weak abelian cube xyz . Let k_0 be the position at which it starts and put

$$k_1 = k_0 + |x|, \quad k_2 = k_1 + |y|, \quad k_3 = k_2 + |z|.$$

All three blocks are nonempty, so $k_0 < k_1 < k_2 < k_3$. There is a vector $f \in \mathbb{Q}_{\geq 0}^4$ such that

$$\psi(x) = |x|f, \quad \psi(y) = |y|f, \quad \psi(z) = |z|f.$$

Its coordinates sum to 1, since $f = \psi(x)/|x|$. If $v = Lf$, then the third coordinate of v is

$$2f_0 + 4f_1 + 6f_2 + 8f_3 \geq 2 > 0.$$

In particular, $v \neq 0$. Applying (10) to the three blocks gives

$$\begin{aligned} P_{N_{k_1}} - P_{N_{k_0}} &= |x|v, \\ P_{N_{k_2}} - P_{N_{k_1}} &= |y|v, \\ P_{N_{k_3}} - P_{N_{k_2}} &= |z|v. \end{aligned}$$

Thus the four distinct vertices $P_{N_{k_0}}, P_{N_{k_1}}, P_{N_{k_2}}, P_{N_{k_3}}$ lie on the line $P_{N_{k_0}} + \mathbb{R}v$. Each associated index is a zero position of $\mathbf{t}$, contradicting Proposition 5. This excludes every weak abelian cube, regardless of its position or its three block lengths. $\square$

Remark. There is also a description using only the zero positions of the four-letter fixed point $\mathbf{t}$. The return lengths associated with output letters 0, 1, 2, 3 are 2, 4, 6, 8, respectively. Hence, with $N_0 < N_1 < \cdots$ enumerating the zero positions of $\mathbf{t}$,

$$w_k = \frac{N_{k+1} - N_k}{2} - 1 \quad (k \in \mathbb{N}).$$

Lemma 2 proves that these gaps are always in $\{2, 4, 6, 8\}$, so this formula is equivalent to the explicit substitution-and-coding construction in Theorem 1.

References

- [1] Stijn Cambie and Erik Kalviainen, *An infinite small-step $\mathbb{Z}^3$ -walk with no collinear triple*, arXiv:2609.01766v1, 2026. <https://arxiv.org/abs/2609.01766>.